DEPARTMENT OF

BIOTECHNOLOGY

VISION OF THE DEPARTMENT

To emerge as a premier Centre in Biotechnology with scientific pursuits and focusing on human values & professional ethics.

MISSION OF THE DEPARTMENT

- Imparting knowledge of the fundamentals of Engineering, Science and Technology in students by providing good academic environment to pursue undergraduate, Post graduate and Doctoral programs in Biotechnology for a successful professional career.
- Developing liaison with Academia, R & D institutions and Biotechnology Industry for exposure in practical aspects in engineering and solution of industry oriented and societal problems, entrepreneurial pursuit and project management.

- Inculcating interpersonal skills, teamwork, professional ethics, IPR and regulatory issues in students to improve their employability in changing global environment.

- Promoting strong research culture in students for life – long learning.

B. TECH.
(BIOTECHNOLOGY)

B.TECH. (BIOTECHNOLOGY)

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: Demonstrate strong foundational knowledge in biotechnology, molecular biology, genetics, and bioprocess engineering to solve real-world problems.

PEO-II: Apply modern tools, computational methods, and experimental techniques to design, optimize, and scale bioproducts and sustainable technologies.

PEO-III: Exhibit the ability to work effectively in multidisciplinary teams while maintaining high ethical standards, biosafety, and regulatory compliance.

PEO-IV: Engage in lifelong learning and demonstrate effective communication skills and professional responsibility to excel in academia, industry, or entrepreneurship in the biotechnology sector.

B.TECH. (BIOTECHNOLOGY)

PROGRAM OUTCOMES

PO-1: **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B.TECH. (BIOTECHNOLOGY)

PROGRAM SPECIFIC OUTCOMES

PSO-1: Apply molecular biology, genetics, bioprocess engineering, and bioinformatics to design, optimize, and scale up bioproducts like vaccines, enzymes, biofuels and therapeutics.

PSO-2: Ability to conduct experiments, analyse, interpret data using modern tools, computational approaches, while contributing to innovation in bioproduct development and sustainable technologies.

PSO-3: Demonstrate the capacity to work in multidisciplinary teams, uphold ethical, biosafety and regulatory standards.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B. TECH. I YEAR

BIOTECHNOLOGY

I SEMESTER R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT104	Basic Mathematics	3	1	0	4	4
25PC1BT101	Fundamentals of Biology					
25BS1CH101	Chemistry for Engineers	3	0	0	3	3
25HS1EN101	English for Skill Enhancement	3	0	0	3	3
25ES1DS101	Problem Solving Through C	3	0	0	3	3
25ES3ME102	Engineering Graphics	0	0	6	6	3
25BS2CH101	Engineering Chemistry Laboratory	0	0	2	2	1
25HS2EN101	English Language and Communication Skills Laboratory	0	0	2	2	1
25ES2DS101	Problem Solving Through C Laboratory	0	0	2	2	1
25ES2ME101	Engineering Workshop	0	0	2	2	1
25MN6HS101	Induction Programme	2	0	0	2	0
Total		**14**	**1**	**14**	**29**	**20**

II SEMESTER R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT109	Mathematics for Biotechnology	3	0	0	3	3
25BS1PH101	Applied Physics	3	0	0	3	3
25ES1EE102	Basic Electrical and Electronics Engineering	3	0	0	3	3
25ES1AD102	Foundations of Python Programming	3	0	0	3	3
25PC1BT102	Cell Biology	3	0	0	3	3
25ES1CE102	Basics of Engineering Mechanics	2	0	0	2	2
25BS2PH101	Applied Physics Laboratory	0	0	2	2	1
25ES2EE102	Basic Electrical and Electronics Engineering Laboratory	0	0	2	2	1
25ES2AD102	Foundations of Python Programming Laboratory	0	0	2	2	1
25MN6HS103	Happiness and Wellbeing	2	0	0	2	0
Total		**19**	**0**	**6**	**25**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1MT104) BASIC MATHEMATICS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics Knowledge at High School Level

COURSE OBJECTIVES:
- To understand different types of matrices, their operations and roots of equations
- To understand the concept of complex numbers
- To understand the concept of trigonometry
- To understand the basic concept of differentiation
- To understand the basic concept of integration

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Find addition and multiplication of two matrices and inverse of non-singular matrix
CO-2: Solve the simple problems related complex numbers
CO-3: Solve the simple problems related trigonometry
CO-4: Solve the simple problems related differentiation
CO-5: Solve the simple problems related integration

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	2	1	2	1	-	-	-	-	-	-	-
CO-2	3	2	1	1	-	-	-	-	-	-	-	-	-	-
CO-3	3	2	1	1	-	-	-	-	-	-	-	-	-	-
CO-4	3	3	2	2	1	1	1	-	-	-	-	-	2	1
CO-5	3	3	2	2	1	1	1	-	-	-	-	-	2	1

UNIT-I:
Matrices: Introduction of matrix, order of matrix, types of matrices, [Row, Column, singleton, zero, Rectangular, Square, Triangular, Diagonal, Scalar, Identity matrices] definitions with example. Addition, Difference, Multiplication of matrices with example problems, determinant, trace, inverse of matrix, definition of special type matrices [symmetry, skew symmetry, orthogonal].

UNIT-II:
Complex Numbers: Introduction of complex numbers, addition, difference, multiplication and division of complex numbers, conjugate modulus of complex numbers, principal argument of complex number, Argand plane and polar representation, de Moivre’s theorem (without proof), nth roots of unity.

UNIT-III:
Trigonometry: Trigonometric ratios with their graphical representation. Compound angles, multiple and sub multiple angles, transformations, periodicity.

UNIT-IV:
Functions and Differentiation: Function: Definition of set, domain, co-domain, range of function, Types of functions, some standard functions and their properties, graphical representation, [polynomial function, modulus function, exponential, logarithmic, trigonometric function] with example problems, odd and even functions.
Differentiation: Definition of intervals, and neighbourhoods, concept of limits, left and right-hand limits, existence of limit, indeterminate forms, standard limits, definition of continuity with example problems, and discontinuity. Derivative rule for the addition, difference, product and division of two functions.

UNIT-V:
Integration: Definition standard integrals, integration by the method of substitution, integration by parts, integration of sum, difference, multiplication of two functions, general properties of definite integral, integration of even and odd functions.

TEXT BOOKS:
1. Mathematics Text Book for Classes XI and XII, NCERT
2. Text Book of Intermediate Mathematics (Paper-IA and IB), Telugu Akademi

REFERENCES:

1. Calculus, Santi Narayana and P. K. Mittal, Differential S. Chand
2. Handbook - Mathematics (Key notes, Terms, Definitions, Formulae), Arihant Publications

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25PC1BT101) FUNDAMENTALS OF BIOLOGY

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basics of Biological Sciences

COURSE OBJECTIVES:
- To get awareness of how the earth and life formation occurred, diversity of life forms, general characteristics used for classification, and benefits and dangers of life forms
- To understand how roots stem and leaves develop classification and importance of plant growth factors and plants
- To study the characteristic differences between chordates and non-chordates with some examples
- To get knowledge about the basic structure and function of different organ systems of human system and their inter-relationship
- To understand the scope and importance of Applied biology in terms of application and their commercial importance

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the composition of ingredients required for life and how they develop and apply the knowledge acquired to evaluate pros and cons of bacteria, Algae, fungi and protozoan for human welfare
CO-2: Distinguish major plant phyla and analyze the importance of growth factors in plant development and commercial values of plants
CO-3: Differentiate the major animal phyla and assess the effects of protozoans and helminthes on human welfare
CO-4: Demarcate the significance of each system and their structure and functional inter relationships between the organ systems
CO-5: Assess the importance of diversity in biology and explore the benefits in terms of commercialization and application in various fields

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	-	-	1	-	1	-	-	1	-	2	-	-	-
CO-2	3	-	1	1	-	1	-	-	1	-	2	1	2	-
CO-3	3	-	1	1	-	1	-	-	1	-	2	1	1	-
CO-4	3	-	1	1	-	1	-	-	1	-	2	1	2	-
CO-5	-	2	2	2	-	1	-	-	1	-	2	3	2	-

UNIT-I:
Introduction to Biology: Origin of life, Diversity in biological systems, Kingdom systems of classification, General characters, brief account on ecology, morphology, nutrition,

locomotion and reproduction. Beneficial and harmful effects of Bacteria, Algae, Fungi and Protozoans with examples.

UNIT-II:
Biology of Plants: Classification of Plant Kingdom, Concepts of Growth, Meristems, Development of different plant organs; Plant Growth Regulators; Economic Importance of Plants, Biology of Pests in relation to Rice, Cotton, and Groundnut, Photosynthesis – overview.

UNIT-III:
Biology of Animals: Classification of Animal Kingdom, General Characters of Chordates and Non-chordates, Protozoan Parasites (Plasmodium, Entamoeba histolytica), Helminth parasites (Taenia solium, Ascaris).

UNIT-IV:
Human Biology: Introduction to human body, Comparison between animal biology and human biology. Digestive, Respiratory, Circulatory, Nervous, Endocrine, Excretory and Reproductive system.

UNIT-V:
Applied Biology: Scope and Importance, Introduction to Drugs and Chemicals from Plants & Animals, Commercially important Enzymes, Vaccines, Antibodies, Cell and Gene therapy, Bacteriophages, Biotherapeutics Biofuels, Bio-fertilizers, Bio-pesticides, Bio-indicators, Biosensors, and Bioremediation.

TEXT BOOKS:

1. Fundamentals of Biology, Haupt Arthur W., McGraw-Hill, 2007
2. Introduction to Biology And Biotechnology, Vaidyanath K., K. Pratap Reddy, 2nd Edition, BS Publications, 2009
3. Fundamentals of Biology: CBSE Class 12 - Set of Textbook & Practice Book, Wiley Editorial (Maestro) Oswal Printers & Publishers, 2019

REFERENCES:

1. Human Physiology, Dr. C. C. Chatterjee, Vol. I and II, 11th Edition, Medical Allied Agency, Kolkata, 1987
2. Biotechnology Volume I & 2, H. G. Rehm and G. Reed, 2012
3. Basic Biotechnology, Colin Ratledge and Bjorn Kristiansen, 2nd Edition, Cambridge University Press, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1CH101) CHEMISTRY FOR ENGINEERS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: General Chemistry and Basic Mathematics

COURSE OBJECTIVES:
- To analyze the quality of water for sustainable living
- To imbibe the conceptual knowledge of electrochemical processes and corrosion science
- To outline the importance of non-conventional energy sources and portable electronic devices
- To acquire knowledge about polymer science and its applications in various fields
- To recognize the significance of analytical techniques and advanced functional materials in engineering, industrial, environmental, and biomedical applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the specification of water regarding its usage in domestic & Industrial scenarios
CO-2: Interpret electrode potentials and predict the suitable corrosion control methods in safeguarding the structures
CO-3: Recognize the transformations in energy sources & battery technology
CO-4: Analyze the efficacy of polymers in diverse applications
CO-5: Identify the use of spectroscopic techniques and advanced functional materials in varied applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	1	1	1	1	1	2	2	2	3
CO-2	3	2	2	1	1	1	1	1	1	1	2	1	1	3
CO-3	3	2	2	1	1	1	1	1	1	1	2	2	2	3
CO-4	3	2	2	1	1	2	1	1	1	1	2	3	3	3
CO-5	3	2	1	1	1	1	1	1	1	1	2	3	3	3

UNIT-I:
Water and its Treatment: Introduction- Hardness, types, degree of hardness and units – numerical Problems. Estimation of hardness of water by complexometric method. Potable water and its specifications (WHO) – Steps involved in the treatment of potable water – screening, sedimentation, coagulation, filtration and disinfection by chlorination and break-point chlorination.
Boiler Troubles: Scales, Sludges, Internal treatment of boiler feed water – Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods – Softening of water by Ion-exchange process. Membrane Separation –

Types of membranes [Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF)], advantages and applications. Desalination of brackish water – Reverse osmosis.

UNIT-II:

Electrochemistry and Corrosion: Introduction – Electrochemical cell & galvanic cell, electrode potential, standard electrode potential, Nernst equation (no derivation). Types of electrodes – Reference electrodes – Construction, working principle, advantages, limitations and applications of Primary reference electrode (Standard Hydrogen Electrode – SHE), Secondary reference electrode (Calomel electrode) and Ion-selective electrode (Combined glass electrode).

Corrosion: Introduction – Definition, causes and effects of corrosion, Theories of corrosion – chemical and electrochemical theories of corrosion. Types of corrosion – galvanic, waterline and pitting corrosion. Factors affecting rate of corrosion – nature of metal (position, passivity, purity, areas of anode and cathode) & nature of environment (temperature, pH, humidity). Corrosion control methods – Cathodic protection Method – Sacrificial anode and impressed current cathodic protection methods.

UNIT-III:

Energy Sources and Battery Technology:

Fuels: Introduction and characteristics of good fuel, classification based on physical state.

Fossil Fuels: Petroleum – Refining of Crude oil, Cracking – Types of cracking – Moving bed catalytic cracking. Synthetic Fuels: Fischer-Tropsch process.

Gaseous Fuels: Composition and uses of LPG, CNG and Hythane. Green Hydrogen (generation by water electrolysis – PEM, Alkaline, Solid Oxide).

Batteries: Introduction – Classification of batteries – Primary, secondary, reserve batteries and fuel cells (differences and examples). Construction, working and applications of Zn-air (primary battery) and Lithium-ion battery (secondary battery), Characteristic features and Advantages of Metal ion batteries (Sodium Ion Battery). Fuel Cells – Construction and applications of Proton Exchange Membrane Fuel Cell (PEMFC).

UNIT-IV:

Polymers: Definition – Classification of polymers – Types of polymerizations – Addition and condensation polymerization, differences between thermoplastics and thermosetting plastics. Plastics, Elastomers and Fibers: Synthesis, properties and applications of Teflon, Buna-S and Kevlar.

Fiber Reinforced Plastics (FRPs): Classification, features and applications.

Conducting Polymers: Definition, classification and applications.

Shape Memory Polymers: Definition, classification and applications.

Biodegradable Polymers: Definition & Example, features and applications of Polylactic acid (PLA).

UNIT-V:

Spectroscopic Applications and Advanced Functional Materials: Interpretative spectroscopic applications of UV-Visible spectroscopy for analysis of pollutants in dye industry, IR spectroscopy in night vision – security, Pollution Under Control – CO sensor (Passive Infrared detection), Raman spectroscopy (application) – Tumour detection in medical applications.

Advanced Functional Materials: Smart materials - Shape Memory Alloys (Example: Nitinol), Nanomaterials – Characterization by STM & AFM.

Self-healing Materials: Definition, working principle and applications.

Characteristic Features of Biosensing Materials: Example: Amperometric Glucose monitoring sensor.

TEXT BOOKS:

1. Engineering Chemistry, P. C. Jain and M. Jain, Dhanpat Rai and Company, 2010
2. Engineering Chemistry, Rama Devi, Dr. P. Aparna and Rath, Cengage Learning, 2025
3. Engineering Chemistry, Thirumala Chary, Laxminarayana & Shashikala, Pearson Publications, 2020

REFERENCES:

1. Engineering Chemistry, Shashi Chawla, Dhanpat Rai and Company, 2011
2. Engineering Chemistry, Shikha Agarwal, Cambridge University Press, 2015
3. Textbook of Engineering Chemistry, Jaya Shree Anireddy, Wiley Publications
4. Engineering Analysis of Smart Material Systems, Donald J. Leo, Wiley, 2007
5. Challenges and Opportunities in Green Hydrogen, Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R. K. Sinha, Springer Nature, 2024

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS1EN101) ENGLISH FOR SKILL ENHANCEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basic English

COURSE OBJECTIVES:
- To enhance vocabulary through word formation processes
- To read and comprehend different kinds of texts
- To write clear, concise paragraphs, essays, and letters to produce appropriate technical prose
- To improve coherence and cohesion in writing and speaking
- To recognize and practice the use of rhetorical elements necessary for the successful practice of scientific and technical communication

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Use vocabulary contextually and effectively in oral and written communication
CO-2: Demonstrate an understanding of the rules of functional Remedial Grammar and sentence structures
CO-3: Demonstrate improved competence in Standard Written English, including Remedial Grammar, sentence and paragraph structure, and coherence, and use this knowledge to accurately communicate technical information
CO-4: Apply principles of writing while developing paragraphs, essays, letters and technical prose
CO-5: Employ appropriate rhetorical patterns of discourse in scientific and technical communication

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	-	1	1	1	2	3	2	1	1	1
CO-2	2	2	2	2	2	2	2	1	3	3	2	1	1	1
CO-3	2	2	2	2	1	3	2	1	3	3	2	1	1	1
CO-4	1	1	1	1	1	2	2	1	2	3	2	1	1	1
CO-5	1	1	1	1	-	2	1	1	2	2	1	1	1	1

UNIT-I:
1. **Reading:** -The Generation Gap' by Benjamin M. Spock
   The Lotos-Eaters – Alfred Tennyson, The Land of High Passes – Bhagyalakshmi Krishnamurthy
2. **Grammar:** Degrees of comparison, Conjunctions, and Prepositions – Identifying Common Errors
3. **Vocabulary:** Word Formation (Affixation, Compounding, Conversion, Blending, Borrowing)

4. **Writing:** Punctuation, Clauses and Sentences, Transitional Devices, Paragraph Writing- Process

UNIT-II:

1. **Reading:** Emerging Technologies
   Medicate – Oleksandr Gorpynich Of Studies – Francis Bacon
2. **Grammar:** Articles Noun-Pronoun Agreement; Concord
3. **Vocabulary:** Word Formation (Prefixes, Suffixes, Root Words)
4. **Writing:** Essay Writing -Descriptive, Argumentative, Expository

UNIT-III:

1. **Reading: Leisure- William Henry Davies -Be Thankful - Unknown Author**
   The Rising of the Moon – Lady Gregory, With The Photographer – Stephen Leacock
2. **Grammar:** Misplaced Modifiers
3. **Vocabulary:** Synonyms and Antonyms
4. **Writing:** Letter Writing- Formal Letter Writing– Letter of Complaint, Letter of Requisition, Email Writing, Email Etiquette

UNIT-IV:

1. **Reading:** -Why a Start-Up Needs to Find its Customers First'- Pranav Jain
   Too dear – Leo Tolstoy, Shooting an Elephant – George Orwell
2. **Grammar:** Clichés, Redundancies
3. **Vocabulary:** Common Abbreviations
4. **Writing:** Summary Writing, Job Application, Resume Writing

UNIT-V:

1. Reading: **'Professional Ethics'**
2. Patterns of Writing: Comparison and Contrast
3. Patterns of Writing: Cause and Effect Paragraphs
4. Patterns of Writing: Classification Paragraph
5. Patterns of Writing: Problem–Solution Pattern of Writing

TEXT BOOKS:

1. Board of Editors, English for the Young in the Digital World, Orient Black Swan, 2025
2. Unlocking English: A Skill-Based Approach for Language Proficiency
3. Technical Communication, Rebecca E. Burnett, 6th Edition, Cengage Learning

REFERENCES:

1. Practical English Usage, Swan Michael, New Edition, Oxford University Press, 2016
2. English Grammar Just for You, Karal, Rajeevan, Oxford University Press, 2023
3. Empowering with Language: Communicative English for Undergraduates, Cengage Learning, 2024
4. Communication Skills – A Workbook, Sanjay Kumar & Pushp Lata, Oxford University Press, 2022
5. Study Writing, Hamp Liz., Lyons and Heasly, Ben, C U P, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1DS101) PROBLEM SOLVING THROUGH C
(Common to AME, and Bio-Technology)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:
- To relate basics of programming language constructs and problem-solving techniques
- To classify and implement control structures and derived data types
- To analyze and develop effective modular programming
- To construct mathematical problems and real time applications using C language

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understanding computer fundamentals, problem-solving tools and implement programs using various data types, operators, and expressions
CO-2: Apply conditional and iterative control structures to solve computational problems
CO-3: Design and implement programs using arrays (1D, 2D) and strings
CO-4: Apply modular programming principles using functions and perform basic file handling operations
CO-5: Implement advanced C constructs for solving real-world problems

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	3	1	-	-	-	-	2	2	1	2	2	1
CO-2	2	2	3	2	1	-	-	-	2	2	-	1	1	2
CO-3	2	2	3	2	1	1	-	-	2	2	-	1	2	2
CO-4	1	2	3	1	1	1	-	-	2	2	2	2	1	1
CO-5	1	2	3	1	1	1	-	-	2	2	2	1	1	2

UNIT-I:
Introduction to Programming: Introduction to components of a computer system (I/O devices, memory, processor). Algorithm, Flow chart with examples. Introduction to the C Language – Programming Language levels, Identifiers, Data Types, Variables, Constants, Input / Output, Operators (Arithmetic, relational, logical, bitwise etc.), Expressions, Precedence and Associativity, Type casting and Type conversion, Simple C Program examples.

UNIT-II:
Conditional Branching and Loops: Statements- Selection Statements (making decisions) – branching with if, if-else, nested if-else, else-if ladder, switch case Repetition statements (loops)-while, for, do-while statements, nested loops, other statements related to looping– break, continue.

UNIT-III:

Arrays: One- and two-dimensional arrays, creating, accessing and manipulating elements of arrays.

Strings: Introduction to strings, handling strings as array of characters, string and character functions available in C, arrays of strings.

UNIT-IV:

Functions: Declaring a function, function types with respect to return values and parameters, call by value, call by reference. Recursion with examples. Some C standard functions and libraries, Storage classes.

UNIT-V:

Pointers: Idea of pointers, defining pointers, use of pointers in self-referential structures, dynamic memory allocation.

Structures & Unions: Defining structures and array of structures, Unions.

Files: File types, character wise reading and writing operations (fopen(), fclose(), fputc(), fgetc()).

TEXT BOOKS:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India
2. Schaum's Outline of Programming with C, Byron Gottfried, McGraw-Hill

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, IV Edition, McGraw-Hill
2. Let Us C, Yashvant Kanetkar, BPB Publications
3. Programming in ANSI C, E. Balaguruswamy, Tata McGraw-Hill

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES3ME102) ENGINEERING GRAPHICS

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE OBJECTIVES:

- To know the importance of engineering scales and curves
- To learn to use the orthographic projections for points, lines and planes in different positions
- To learn to draw the projections of solids and its sections in different positions
- To learn to draw the development of surfaces and intersection of solids
- To learn to draw the isometric projections of solids

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply the concepts of scales and engineering curves in construction using AutoCAD

CO-2: Solve the problem of projections of points, lines and planes in different positions using AutoCAD

CO-3: Draw the orthographic projections of solids and its sections in different positions using AutoCAD

CO-4: Obtain the development of surfaces of solids and intersection of solids using AutoCAD

CO-5: Construct the isometric projections of solids using AutoCAD

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	-	-	3	2	-	-	-	-	2	-	1	-
CO-2	3	2	-	-	3	2	-	-	-	-	2	-	1	-
CO-3	3	2	-	-	3	2	-	-	-	-	2	-	1	-
CO-4	3	2	-	-	3	2	-	-	-	-	2	-	1	-
CO-5	3	2	-	-	3	2	-	-	-	-	2	-	1	-

Introduction to AutoCAD Software: User interface, Menu System, Toolbars (Draw, Modify, Dimension, Layers), Setting drawing area, Status Bar (ortho, grid, snap, osnap, iso, lineweight), Display control commands (pan, zoom), Print setup.

UNIT-I:

Introduction to Engineering Graphics: Principles of Engineering Graphics and their Significance, Conventions, Drawing instruments.

Engineering Curves

Conic Sections: Ellipse, Parabola and Hyperbola- General methods only, Rectangular Hyperbola

Cycloidal Curves & Involutes: Cycloid, Epicycloid, Hypocycloid and Involutes
Scales: Plain, Diagonal and Vernier Scales

UNIT-II:
Orthographic Projections: Principles of Orthographic Projections, Conventions, Projections of Points in all positions; Projections of lines and planes inclined to both the planes - Auxiliary Views

UNIT-III:
Projections of Regular Solids: Projections of regular Solids like Prism, Cylinder, Pyramid and Cone inclined to both the Planes - Auxiliary Views
Sections of Solids: Section and sectional views of right regular solids like Prism, Cylinder, Pyramid and Cone – Auxiliary Views

UNIT-IV:
Development of Surfaces of Right Regular Solids: Development of surfaces of Right Regular Solids like Prism, Pyramid, Cylinder and Cone
Intersection of Solids: Intersection of prism vs prism, cylinder vs cylinder

UNIT-V:
Isometric Projections: Principles of isometric projection, Isometric Scale, Isometric Views, Conventions, Isometric Views of lines, Planes and Solids like Prism, Pyramid, Cylinder and Cone
Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions.

TEXT BOOKS:
1. Engineering Drawing, N. D. Bhatt, 53rd Edition, Charotar Publishing House, 2016
2. Textbook on Engineering Drawing, K. L. Narayana & P. Kannaiah, Scitech Publishers, 2010
3. Engineering Drawing and Computer Graphics, M. B. Shah & B. C. Rana, Pearson Education, 2010

REFERENCES:
1. Engineering Drawing and Graphics using AutoCAD, T. Jeyapoovan, 3rd Edition, S. Chand and Company Ltd.
2. Engineering Drawing, Basant Agrawal and C. M. Agrawal, 3rd Editon, McGraw-Hill
3. Mastering AutoCAD 2021 and AutoCAD LT 2021, George Omura and Brian C. Benton (AutoCAD 2021), 1st Edition, John Wiley & Sons

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/112103019
2. http://vlabs.iitb.ac.in/vlabs-dev/labs/mit_bootcamp/egraphics_lab/labs/index.php

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS2CH101) ENGINEERING CHEMISTRY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Knowledge of Volumetric Analysis and Mathematics

COURSE OBJECTIVES:

- To understand the preparation of standard solutions and handling of instruments
- To determine and evaluate the water quality
- To measure physical properties like absorption of light, surface tension, pH, conductance, and viscosity of various liquids
- To conduct and collect the experimental data using different laboratory techniques
- To summarize the data and find the applicability to real world scenario

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Learn and apply the basic laboratory methodologies for the preparation of the standard solutions and handling of instruments
CO-2: Estimate the ions / metal ions present in domestic and industrial water
CO-3: Utilize the instrumental techniques to assess the physical properties of oils and water
CO-4: Analyze the experimental data to predict solutions for complex engineering problems
CO-5: Apply the skills gained to address societal issues related to real world scenario

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	1	1	1	1	1	2	2	2	2
CO-2	3	2	2	2	1	2	1	1	1	1	2	1	2	1
CO-3	3	2	1	2	1	2	1	1	1	1	2	2	2	1
CO-4	3	2	2	2	1	2	1	1	1	1	2	2	2	1
CO-5	3	2	2	2	1	2	1	1	1	1	2	2	2	2

LIST OF EXPERIMENTS:

VOLUMETRIC ANALYSIS:

1. Estimation of hardness of water by complexometric method using EDTA.
2. Determination of chloride content in the given sample water using Argentometric method.
3. Determination of Iron using Potassium Dichromate.

COLORIMETRY:

4. Estimation of copper present in the given solution by colorimetric method.

CONDUCTOMETRY:

5. Conductometric titration of Acid vs Base.
6. Conductometric titration of mixture of strong acid and weak acid by strong base.

pH METRY:

7. Titration of Acid vs Base using pH metric method.
8. Determination of viscosity of sample oil by Redwood Viscometer-I.
9. Estimation of acid value of given lubricant oil.
10. Determination of surface tension of a liquid by drop method using Stalagmometer.

PREPARATIONS:

11. Preparation of Bakelite.
12. Preparation of Nylon-6,6.
13. Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.
14. **VIRTUAL LAB EXPERIMENTS / CBP:**

- Construction of advanced Fuel cell and it's working.
- Smart materials for Biomedical applications
- Batteries for electrical vehicles.
- Functioning of solar cell (materials used) and its applications.

TEXT BOOKS:

1. Laboratory Manual on Engineering Chemistry, S. K. Bhasin and Sudha Rani, Dhanpat Rai Publications
2. College Practical Chemistry, V. K. Ahluwalia, Sunitha Dhingra, Adargh Gulati, University Press
3. Practical Chemistry, Dr. O. P. Pandey, D. N. Bajpai, and Dr. S. Giri, S. Chand

REFERENCES:

1. Vogel's Textbook of Quantitative Chemical Analysis, G. N. Jeffery, J. Bassett, J. Mendham and R. C. Denny, Longmann, ELBS
2. Advanced Practical Physical Chemistry, J. D. Yadav, Goel Publishing House
3. Practical Physical Chemistry, B. D. Khosla, R. Chand and Sons
4. Lab Manual for Engineering Chemistry, B. Ramadevi and P. Aparna, S. Chand, 2022
5. Vogel's Textbook of Practical Organic Chemistry, 5th Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS2EN101) ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

INTRODUCTION:
The aim of the English Language and Communication Skills Lab (ELCS Lab) is to develop the communicative skills of learners to function productively in academic and professional settings. This course focuses on developing Listening Skills, Reading Skills, Speaking Skills, Information transfer, Vocabulary, and pronunciation.

Professional students need reading strategies to explore scientific information and process the information. Further, they need to present and share this information in speaking and/or writing with their peers and scientific community. In addition to these, they also need good vocabulary for the presentation of their ideas. Practice is given not only in processing information to identify main ideas and relevant facts, to compare and contrast information and finally draw conclusions but also in presenting the same in speaking and writing.

COURSE OBJECTIVES:
- To train students to use neutral accent through phonetic sounds, symbols, stress and intonation
- To provide practice in vocabulary usage & grammatical construction
- To provide ample practice in LSRW skills and train the students in oral presentations, public speaking, role play, and situational dialogue
- To provide practice in defining technical terms and describing processes
- To equip students with excellent writing skills and information transfer skills

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Speak fluently with a neutral accent
CO-2: Use contextually apt vocabulary and sentence structures
CO-3: Make Presentations with great confidence
CO-4: Define technical terms and describe processes
CO-5: Write accurately, coherently, and lucidly

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	2	2	2	-	-	-	2	3	-	2	-	-	-
CO-2	1	2	2	2	3	3	2	2	3	2	2	-	-	-
CO-3	-	1	2	1	-	1	2	2	3	2	2	-	-	-
CO-4	2	2	2	2	2	-	-	1	3	2	2	-	-	-
CO-5	1	2	1	2	2	2	2	1	3	3	3	-	-	-

LIST OF EXPERIMENTS:

UNIT-I:

CALL

1. Common errors with Speech Sounds – Identify elements of Mother Tongue Influence,
2. Listening Skills – Types of Listening- Barriers – Active Listening Practice
3. Reading Skills -Skimming & Scanning-Reading for Gist -Reading for Specific Details

Oral Communication Lab:

1. Non-Verbal Communication
2. Formal and Informal English, Small Talk
3. Greetings – Introducing oneself & others - Elevator Pitch

UNIT-II:

CALL

1. **Activity:** Listening for information
2. Listening comprehension
3. **Reading Skills:** Reading comprehension-contextual vocabulary

Oral Communication Lab

1. Participatory Conversation
2. Role Playing in Formal Situations, Situational Dialogues – Two-Way (E. g. Elevator Pitch)
3. Media Etiquette, Conference Etiquette

UNIT-III:

CALL

1. **Phonetics:** Errors in Pronunciation, Neutralizing Mother Tongue Influence
2. **Listening Skills:** Listening for Gist
3. **Reading Skills:** SQ3R Method, Inferring

Oral Communication Lab

1. Describing Objects / Situations/ Places, People, Events,
2. Process Description

UNIT-IV:

CALL

1. **Listening Skills:** Techniques for Listening –Listening for Specific Details
2. **Reading Skills:** Making inferences

Oral Communication Lab

1. Story Telling; Story STAR, Sequencing, Creativity
2. Telling and Retelling Stories- Collage

UNIT-V:

CALL

1. **Listening Skills:** Listening for Evaluation, Writing the Summary
2. **Prompt Engineering:** Generating Appropriate Prompts, Keyword Extraction
3. Definition of a Technical Term; System Flow

Oral Communication Lab

1. Report Presentation
2. Short Presentation

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2DS101) PROBLEM SOLVING THROUGH C LABORATORY

(Common to AME and Bio-Technology)

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To acquire hands-on experience in problem solving using C language
- To develop modular, efficient and readable programs using control structures, arrays, strings, and functions
- To learn concepts of recursion, pointers, structures, and unions
- To strengthen problem-solving skills with file handling, dynamic memory allocation

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply data types, operators, and control structures to solve computational problems
CO-2: Develop modular programs using functions and recursion
CO-3: Implement programs using arrays, strings, pointers, structures, and unions
CO-4: Solve real-time problems using file handling and dynamic memory concepts

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	3	2	1	-	-	-	2	2	-	1	1	1
CO-2	2	2	3	2	1	-	-	-	2	2	-	1	1	1
CO-3	1	2	3	1	1	1	-	-	2	2	2	1	1	1
CO-4	1	2	3	1	1	1	-	-	2	2	2	1	1	1
CO-5	2	2	3	2	1	1	-	-	2	2	2	1	1	1

LIST OF EXPERIMENTS:

WEEK 1:

- Programs on input/output statements (scanf, printf).
- Programs on arithmetic operators (sum, difference, product, quotient).
- Programs on expression evaluation, operator precedence, type conversion, and casting.

WEEK 2:

- Program to check whether a number is even or odd using if-else.
- Program to find the largest of three numbers using nested if-else.
- Program to implement a simple calculator using switch-case.

WEEK 3:

- Program to print multiplication table of a given number using for.

- Program to display sum of first N natural numbers using while.
- Program to compute factorial of a number using do-while.

WEEK 4:
- Program to print patterns (triangles, pyramids) using nested loops.
- Program to demonstrate break, continue, and goto.

WEEK 5:
- Program to read and find sum and average of elements in an array.
- Program to search for an element using linear and binary search.
- Program to sort elements using bubble, selection, and insertion sort.

WEEK 6:
- Program to perform addition of two matrices.
- Program to multiply two matrices.
- Program to find the transpose of a matrix.

WEEK 7:
- Program to check whether a given string is palindrome.
- Program to search for a substring in a given string.
- Programs using string handling functions (strlen, strcpy, strcmp, strcat).

WEEK 8:
- Lab Internal – I

WEEK 9:
- Program to swap two numbers using call by value and call by reference.
- Program to generate Fibonacci series using recursion.
- Program to solve Tower of Hanoi problem using recursion.

WEEK 10:
- Program to store and display student details using structures.
- Program using array of structures (e. g., employee records).
- Program using unions (demonstrating memory sharing).

WEEK 11:
- Program to demonstrate pointers to variables and arrays (1-D and 2-D).
- Program using pointers to strings.

WEEK 12:
- Program to allocate memory dynamically using malloc, calloc, and release with free.

WEEK 13:
- Program to write student records into a file.
- Program to read and display student records from a file.

WEEK 14:
Lab Internal – II

TEXT BOOKS:
1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India

2. Schaum's Outline of Programming with C, Byron Gottfried, McGraw-Hill

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill
2. Let Us C, Yashvant Kanetkar, BPB Publications
3. Programming in ANSI C, E. Balaguruswamy, Tata McGraw-Hill

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2ME101) ENGINEERING WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:
- To know the different popular manufacturing process
- To gain a good basic working knowledge required for the production of various engineering products
- To provide hands on experience about use of different engineering materials, tools, equipment and processes those are common in the engineering field
- To identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand various types of manufacturing processes
CO-2: Fabricate/make components from wood and steels through hands on experience
CO-3: Understand different machining processes like turning, drilling, tapping, etc.
CO-4: Understand electrical and electronic components and their assembly

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	-	1	-	-	-	-	3	3	2	-	-	1	2
CO-2	-	-	1	-	-	-	-	3	3	2	-	-	1	2
CO-3	-	-	1	-	-	-	-	3	3	2	-	-	1	2
CO-4	-	-	1	-	-	-	-	3	3	2	-	-	1	2

LIST OF EXPERIMENTS:

1. TRADES FOR EXERCISES:
i) **Carpentry:** Cross lap joint, Mortise & tenon joint
ii) **Fitting:** L-fitting, Square fitting
iii) **Tin smithy:** Rectangular tray, Rectangular scoop
iv) **Arc welding:** Lap joint, Butt joint
v) **House Wiring:** Single lamp connection, Staircase connection
vi) **Black smithy:** Round to Square, U-Hook
vii) **Machine shop:** Step turning on lathe, Drilling & tapping

2. TRADES FOR DEMONSTRATION:
Plumbing, 3D printing, Power tools in construction and wood working.

TEXT BOOKS:

1. Workshop Manual, P. Kannaiah and K. L. Narayana, 3rd Edition, Scitech, 2015
2. Elements of Workshop Technology Vol. 1 & 2, S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, 13th Edition, Media Promoters & Publishers, 2010

REFERENCES:

1. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Education, 2002
2. Manufacturing Technology-I, S. Gowri, P. Hariharan and A. Suresh Babu, Pearson Education, 2008
3. Processes and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice Hall India, 1998
4. Manufacturing Technology Vol-1 & 2, P. N. Rao, Tata McGraw-Hill, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT109) MATHEMATICS FOR BIOTECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn concept of a rank of the matrix and applying this concept to know the consistency and solving the system of linear equations
- To learn concept of Eigenvalues and Eigenvectors
- To learn concept of finding maxima and minima of function of two variables
- To learn methods of solving the ordinary differential equations of first order
- To learn methods of solving the ordinary differential equations of higher order

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply matrix techniques to solve system of linear equations
CO-2: Find the Eigenvalues and Eigenvectors
CO-3: Find maxima and minima for functions of two variables
CO-4: Solve First Order Ordinary differential equations by analytical methods
CO-5: Solve Higher Order Ordinary differential equations by analytical methods

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	1	2	1	-	-	-	-	-	2	3	-
CO-2	3	3	1	2	2	1	-	-	-	-	-	3	3	-
CO-3	3	3	2	2	1	1	-	-	-	-	-	2	3	-
CO-4	3	2	1	1	1	1	-	-	-	-	-	2	3	-
CO-5	3	2	2	2	1	1	-	-	-	-	-	2	1	-

UNIT-I:
Matrices and System of Linear Equations: Matrices: Rank of a matrix by Echelon form. System of linear equations: solving system of homogeneous and Non-homogeneous equations by Gauss-Elimination method.

UNIT-II:
Eigenvalues and Eigenvectors: Eigenvalues and Eigenvectors and their properties (without proof), Diagonalization of a matrix, Cayley- Hamilton Theorem (without proof), finding inverse and powers of a matrix by Cayley-Hamilton Theorem.

UNIT-III:
Functions of Several Variables: Partial differentiation, Total derivative, Jacobian, Functional dependence, Maxima and Minima of functions of two variables.

UNIT-IV:

First Order Ordinary Differential Equation and Applications:

Differential Equation: order and degree of a differential equation, formation of differential equation, various methods to find general solutions of first order and first-degree differential equation [variable separable, homogeneous, Non – Homogeneous], linear equations. Application: Law of natural growth and decay

UNIT-V:

Higher Order Ordinary Differential Equations: Second and higher order linear differential equations with constant coefficients, Non-Homogeneous terms of the type e^{ax}, Polynomials in x.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 42nd Edition, Khanna Publishers
2. Higher Engineering mathematics, B. V. Ramana, 11th Reprint, Tata McGraw-Hill, 2010

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, John Wiley & Sons, 2006
2. A Textbook of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, Reprint, 2008
3. Linear Algebra: A Modern Introduction, D. Poole, 2nd Edition, Brooks/Cole, 2005
4. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Editon, Narosa Publications, 2016
5. Pharmaceutical Mathematics with Application to Pharmacy, D. H. Panchaksharappa Gowda

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1PH101) APPLIED PHYSICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:

- To explore the working and applications of lasers and fibre optics in modern technology
- To study crystal structures and defects in materials
- To learn nano materials and characterization techniques like XRD & SEM
- To understand the properties and applications of dielectric and magnetic materials
- To identify the importance of quantum mechanics and quantum computing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the principles of lasers and fibre optics and their applications in communication and sensing
CO-2: Understand different types of crystals and the importance of crystal defects
CO-3: Identify the role of nanomaterials in science and engineering applications
CO-4: Classify the dielectric and magnetic materials to illustrate their applications
CO-5: Realize the role of quantum mechanics in quantum computing

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	2	1	-	-	-	1	2	-	-
CO-2	3	3	-	-	-	-	1	-	-	-	1	1	-	-
CO-3	3	2	2	1	1	2	1	-	-	-	1	-	-	-
CO-4	3	3	1	1	1	2	1	-	-	-	1	1	-	-
CO-5	3	3	1	1	1	2	1	-	-	-	1	-	-	-

UNIT-I:

Laser: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, Semiconductor laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Fibre Optics: Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

UNIT-II:

Crystallography & Defects: Introduction, Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance; Introduction to defects, Point defects (Vacancies, Interstitial and Impurities) Line imperfections, Edge and Screw dislocation, Burger vector, Surface defects and volume defects (Qualitative Treatment).

UNIT-III:

Nano-materials & Characterization: Concept of Nano-materials: Surface to volume ratio and quantum confinement, Fabrication: Sol-Gel, Ball milling, Applications. **Characterization:** X -ray diffraction, Bragg's law, powder method, Calculation of average crystallite size using Debye Scherrer's formula, scanning electron microscopy (SEM): block diagram, working principle.

UNIT-IV:

Dielectric Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

Magnetic Materials: Introduction to magnetic materials, classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

UNIT-V:

Quantum Mechanics: Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Quantum Computing: Introduction, concept of quantum computer, advantages of quantum computing over classical computation. classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, Dirac's Bra and Ket notation and their properties, Introduction to entanglement.

TEXT BOOKS:

1. Textbook of Engineering Physics, M. N. Avadhanulu, P. G. Kshirsagar & T. V. S. Arun Murthy A., 11th Edition, S. Chand, 2019
2. Introduction to Solid State Physics, Charles Kittel, John Wiley & Sons
3. Introduction to Classical and Quantum Computing, Thomas G. Wong, Rooted Grove

REFERENCES:

1. Engineering Physics, B. K. Pandey and S. Chaturvedi, 2nd Edition, Cengage Learning, 2022
2. Energy Materials a Short Introduction to Functional Materials for Energy Conversion and Storage, A. S. Bandarenka, CRC Press, Taylor & Francis Group Energy Materials, Taylor & Francis Group, 1st Edition, 2022
3. Concepts of Modern Physics, Aurther Beiser, McGraw-Hill
4. Engineering Physics, P. K. Palanisamy, SciTech Publications

5. Quantum Computing, Jozef Gruska, McGraw-Hill

ONLINE RESOURCES:

1. https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf
2. https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
3. https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf
4. https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1EE102) BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:
- To understand the use of electrical energy in different engineering fields
- To analyze electrical circuits using different network reduction techniques
- To know the working, construction of electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Appreciate the role of electrical energy in various engineering branches and analyze the DC circuits using various network reduction techniques
CO-2: Analyze the various electrical parameters of AC circuits with R-L-C elements
CO-3: Understand the operation of various Electrical Machines & Installations
CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	2	1	-	-	-	-	-	-	2	2	1
CO-2	2	3	2	2	1	-	-	-	-	-	-	2	2	1
CO-3	2	3	2	1	1	-	-	-	-	-	-	2	2	1
CO-4	2	2	2	1	1	-	-	-	-	-	-	2	2	1

UNIT-I:
Introduction to Electrical Energy and DC Circuits: The role of Electrical Energy in modern life and various engineering branches- Circuit Concept – Types of Elements R-L-C parameters – Voltage and Current sources – Independent and dependent sources- Kirchhoff's laws – network reduction techniques: series, parallel, series parallel, star/delta transformations, Superposition theorem, Thevenin's theorem

UNIT-II:
Steady State AC Circuits: Representation of sinusoidal waveforms, average and RMS values, form factor and peak factor, phasor representation, Analysis of single-phase AC circuits consisting of R, L, C, series RL, RC, RLC combinations, real power, reactive power, apparent power, power factor - Three-phase balanced circuits, voltage and current relations in star and delta connections (Derivation only).

UNIT-III:

Electrical Machines: Transformer principle, types, Equivalent circuit, Regulation and Efficiency, DC generator principle, Emf equation, DC motor principle, Back emf, Three phase induction motor, types, principle, torque Slip characteristics, working principle of Synchronous generator, principle of operation of synchronous motor, stepper motor principle and applications.

UNIT-IV:

Diode and BJT: PN Junction Diode: Operation and Symbol, V-I Characteristics, Half Wave Rectifier, Full-Wave Rectifier (Centre Tap and Bridge Rectifiers), Zener Diode, V-I Characteristics

Bipolar Junction Transistor (BJT): Construction (NPN and PNP transistors), Transistor configurations (CC, CB and CE), Transistor as an Amplifier

UNIT-V:

Op-Amp and Potentiometers:

Operational Amplifier (IC 741): Basic information of Op-Amp, Pin Configuration of IC 741, Characteristics of Ideal Op-Amp, Block Diagram of Op-Amp, Applications of Op-Amp (only over view): Inverting and Non- Inverting Amplifier.

Potentiometer (Variable Resistor-Pot): Introduction to potentiometer, definition and working principle, symbol and circuit representation, Types of potentiometers (only over view)-Rotary, Linear and Digital.

TEXT BOOKS:

1. Basics of Electrical Energy for Engineers, V. Ramesh Babu, BS Publications, 2023
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Op-Amps Linear Integrated Circuits, 4th Edition, Ramakant A. Gayakwad, Prentice Hall of India, 2015

REFERENCES:

1. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013
2. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. The potentiometer Hand Book, User Guide to Cost Effective Applications

ONLINE RESOURCE:

1. https://nptel.ac.in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1AD102) FOUNDATIONS OF PYTHON PROGRAMMING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To learn about Python language syntax, semantics, and the runtime environment
- To understand universal computer programming concepts like data types, containers
- To gain knowledge of general computer programming like control structures and functions
- To be well versed with general coding techniques and object-oriented programming

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the fundamental concepts of Python to develop simple programs
CO-2: Apply conditional statements, looping & built-in data structures for efficient problem-solving
CO-3: Develop modular programs, modules, and packages to improve code reusability & maintainability
CO-4: Design and implement Python using Oops principles to model real-world problems
CO-5: Implement error-handling, file i/o & basic data analysis to develop robust & data-driven applications

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	-	-	-	-	3	-	2	2	3
CO-2	3	2	3	2	2	-	-	-	-	-	-	2	2	3
CO-3	3	2	3	2	2	-	-	3	-	-	3	3	2	2
CO-4	3	2	3	2	2	-	-	-	-	3	-	3	2	2
CO-5	3	2	3	2	3	-	-	-	2	-	-	2	2	2

UNIT-I:

Python Fundamentals and Basics: Introduction to Python: History, features, applications. Setting up the environment: Installation, IDEs, basic command-line usage. Variables, Data types (integers, floats, strings, Booleans), and type conversion. Operators: Arithmetic, relational, logical, assignment, bitwise. Basic input/output operations.

UNIT-II:

Control Flow and Data Structures: Conditional statements – if, elif, else. Looping constructs – for loops, while loops, break, continue, pass. Strings – String methods, formatting. List – Creation, manipulation, methods, list comprehensions. Tuples –

Creation, immutability. Sets – Creation, operations. Dictionaries – Creation, key-value pairs, methods.

UNIT-III:
Functions, Modules, and Packages: Function arguments – Positional, keyword, default, variable-length. Scope of variables (local, global), recursion. Introduction to modules – Importing and using built-in and user-defined modules. Introduction to packages.

UNIT-IV:
Object-Oriented Programming (OOP) in Python: Introduction to OOP concepts: Classes, objects, encapsulation, inheritance, polymorphism. Defining classes and creating objects. _init_ constructor. Instance variables and methods. Types of inheritance – Single, multiple, multilevel. Method overriding.

UNIT-V:
Error Handling, File I/O, and Advanced Topics: Exception handling – try, except, else, finally. Raising custom exceptions. File input/output – Reading from and writing to text files. Working with different file modes. Introduction to advanced topics – regular expressions, NumPy, Pandas.

TEXT BOOKS:
1. Fundamentals of Python First Programs, Kenneth. A. Lambert, Cengage
2. Python Programming: A Modern Approach, Vamsi Kurama, Pearson

REFERENCES:
1. Introduction to Python Programming, Gowrishankar. S, Veena A, CRC Press
2. Introduction to Programming Using Python, Y. Daniel Liang, Pearson

ONLINE RESOURCES:
1. https://www.tutorialspoint.com/python3/python_tutorial.pdf
2. https://onlinecourses.nptel.ac.in/noc21_cs78/preview
3. https://www.coursera.org/learn/python?specialization=python#syllabus

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25PC1BT102) CELL BIOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Fundamentals of Cell Structure and Functions

COURSE OBJECTIVES:

- To study the approaches used in discovery of cell, and Cell constituents / organelles, and remembering their structure and functions
- To explain the structure and function of cell organelles within the cell (animal, Plant, Prokaryotes) and the differences which are unique to each cell type
- To get awareness of different membrane transport systems associated with movement of materials from outside into the cell and within the cell between different organelles
- To understand the importance of cell-cell interactions and the consequences of its failure. Understand stages of cell cycle based on DNA ploidy, stages of normal cell division and effects of abnormal / uncontrolled division
- To highlight the importance of the cell-cell communication methods and differences based on signaling molecules, role of receptors and the signaling cascade in cytoplasm and nucleus

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Differentiate the approaches used in the discovery of cell, and Cell constituents / organelles based on their structure and functions
CO-2: Identify and compare cell organelles in terms of structure, function within the cell (animal, Plant, Prokaryotes) and between the cells unique to each cell type
CO-3: Demarcate different membrane transport systems associated with movement of materials from outside the cell and within the cell between different organelles
CO-4: Assess the importance of cell-cell interaction and the consequences of its failure. Differentiate between stages of cell cycle based on DNA ploidy, stages of normal cell division and effects of abnormal / uncontrolled division
CO-5: Analyze the cell-cell communication differences based on signaling molecules, role of receptors and the signaling cascade in cytoplasm and nucleus

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	-	-	-	-	-	-	-	1	-	2	1	-	-
CO-2	3	-	-	-	-	-	-	-	1	-	2	2	1	-
CO-3	3	-	2	-	-	-	-	-	1	-	2	2	1	-
CO-4	3	-	2	3	-	-	-	-	1	-	2	2	1	-
CO-5	3	-	3	3	1	1	-	-	1	-	2	3	2	-

UNIT-I:
Introduction to Cell: Basic properties of cells; Cell theory; Cell complexity – Cell size & shape; Chemistry of the cell, Plasma membrane – structure and function; Cytoplasm; Cytoskeleton – Microtubules, Microfilaments & Intermediate filaments, cell motility – cilia & flagella, Extracellular Matrix (ECM) and its importance. Cell death – Apoptosis and Necrosis.

UNIT-II:
Cell Organelles: Structure and functions of Nucleus, Endoplasmic Reticulum, Golgi complex, Lysosomes, Peroxisomes, Chloroplast & Mitochondria.

UNIT-III:
Membrane Transport / Systems: Passive and Active Transport, Uniport, Symport, Antiport, Permeases, P- Type & V- Type Pumps, Na+/K+ ATPase, Lysosomal & Vacuolar membrane ATP dependent Proton Pumps, Trafficking: Protein Glycosylation (Modifications), Intracellular Protein traffic & targeting, Endocytosis and Exocytosis, Transport in Prokaryotic Cells.

UNIT-IV:
Cell-Cell interactions & Cell Cycle Regulation: Introduction to cell-cell interactions – Cell adhesion molecules: Cadherin's, Ig-like CAMs, Selectins, Integrin's, and cell-cell junctions: Occluding, Anchoring, Communicating junctions.
Cell Cycle Regulation: Cell cycle, Cell Cycle Control & Checkpoints, Interphase, Mitosis, Meiosis & Cytokinesis, Animal Cell Division, and Introduction to Characteristics of Cancer cells, Benign & Malignant tumor, Metastasis. General Characteristics of Cell Differentiation.

UNIT-V:
Receptors & Signal Transduction: Introduction to Intracellular signaling, types of signal receptors - Cytosolic, Nuclear & Membrane bound receptors, Chemo receptors of Bacteria (Attractants & Repellents), Signal Transduction by hormones - Steroid / Peptide hormones; Concept of Secondary messengers, cAMP, cGMP, Protein Kinases, G Proteins; Receptors & Non - receptors associated tyrosine Kinases.

TEXT BOOKS:

1. Molecular Biology of the Cell, Bruce Alberts, https://www.ncbi.nlm.nih.gov/books/NBK20684/
2. Cell and Molecular Biology, De Robertis and De Robertis, Waverly, 1998
3. The Cell: A Molecular Approach, Geoffrey M. Cooper, 2018
4. Cell Biology: Fundamentals and Applications, Arunima Murkerjee, 2009

REFERENCES:

1. Cell & Molecular Biology, Gerald Karp, 2nd Edition, Wiley
2. The World of the Cell, Becker, Reece, Poenie, 3rd Edition, Benjamin Publishers
3. The Biochemistry of Cell Signaling, Ernst J. M. Helmreich, Oxford Press, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1CE102) BASICS OF ENGINEERING MECHANICS

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Matrices and Calculus, Applied Physics

COURSE OBJECTIVES:
- To determine the resultant of coplanar concurrent and non-concurrent forces
- To state and apply the laws of friction
- To find the centroids of lines, areas and volumes
- To find the moment of inertia of areas

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the systems using equilibrium conditions and apply the concepts of mechanics to engineering applications
CO-2: Apply the equations of equilibrium and analyze the bodies lying on rough frictional planes
CO-3: Determine the centroid of built-up sections
CO-4: Determine the moment of inertia of built-up sections

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	-	-	-	-	-	2	-	1	-
CO-2	2	2	1	1	1	-	-	-	-	-	1	-	1	-
CO-3	1	2	1	1	1	-	-	-	-	-	1	-	1	-
CO-4	1	2	1	1	1	-	-	-	-	-	1	-	1	-

UNIT-I:
Forces: Introduction to Engineering Mechanics – Basic concepts - Classification of a force system - Parallelogram law of forces - Triangle law of forces - Polygon law of forces – Law of transmissibility of forces – Principle of superposition – Lami's theorem - Free Body Diagram – Resultant – Equilibrant - Resultant of coplanar concurrent forces - Equations of Equilibrium of Coplanar Systems.

UNIT-II:
Moments: Moment of a force – Varignon's principle - Parallel forces - Resultant of parallel forces – Couple - Moment of a couple about any point lying in the plane - Resolution of a force into a force-couple and vice-versa - Resultant of coplanar non-concurrent forces – Beams – Degree of static indeterminacy - Determination of support reactions - Cantilever beams, Simply supported beams, Over hanging beams subjected to concentrated loads, moments, uniformly distributed loads and uniformly varying loads.

UNIT-III:
Friction: Types of Friction - Laws of friction - Equilibrium of bodies on rough horizontal and inclined planes - Equilibrium of connected bodies on rough horizontal and inclined planes - Ladder friction.

UNIT-IV:
Centroid and Centre of Gravity: Introduction - Centroid - Centroids of lines, areas and volumes – Application to the centroids of built-up sections - Centre of gravity of bodies.

UNIT-V:
Area Moment of Inertia: Introduction - Inertia - Inertia of areas - Radius of gyration - Polar moment of inertia - Parallel axis theorem – Perpendicular axis theorem - Moment of inertia of standard sections – Application to the moment of inertia of built-up sections

TEXT BOOKS:

1. Engineering Mechanics, S. Timoshenko, D. H. Young, J. V. Rao and Sukumar Pati, 5th Edition, McGraw-Hill, 2017
2. Engineering Mechanics: Statics and Dynamics, A. K. Tayal, 14th Edition, Umesh Publications, 2019
3. Engineering Mechanics, S. S. Bhavikatti, 8th Edition, New Age International Publishers, 2021

REFERENCES:

1. Singers Engineering Mechanics: Statics and Dynamics, K. Vijaya Kumar Reddy & J. Suresh Kumar, 3rd Edition, B. S. Publications, 2024
2. Engineering Mechanics: Statics, R. C. Hibbeler, 15th Edition, Pearson Publications, 2021
3. Problems and Solutions in Engineering Mechanics, S. S. Bhavikatti & A. Vittal Hedge, 2nd Edition, New Age International, 2016
4. A Text Book of Engineering Mechanics, Dr. R. K. Bansal, 8th Edition, Laxmi Publications, 2011

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/112/106/112106286/
2. https://nptel.ac.in/courses/112103108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS2PH101) APPLIED PHYSICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: - -

COURSE OBJECTIVES:
- To explore the working principle of lasers and optical fibers through hands-on experiments
- To analyze the characteristics of semiconductors and devices
- To study resonance and magnetic induction in electromagnetic systems
- To synthesize the magnetic materials and study magnetic, dielectric properties of materials
- To develop skills in data analysis, interpretation, and scientific reporting

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Verify laser and optical fibre parameters through experimental study
CO-2: Characterize semiconductors and devices such as solar cell, laser diode etc
CO-3: Demonstrate resonance phenomena and realize tangent law of magnetism
CO-4: Understand the magnetic and dielectric properties of materials
CO-5: Apply scientific methods for accurate data collection, analysis, and technical report writing

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2	-

LIST OF EXPERIMENTS:
Note: Any 8 experiments are to be performed.

1.
a) Determination of wavelength of a laser using diffraction grating.
b) Study of V-I & L-I characteristics of a given laser diode.
2.
a) Determination of numerical aperture of a given optical fibre
b) Determination of bending losses of a given optical fibre.
3. Synthesis of magnetite (Fe_3O_4) powder using sol-gel method.

4. V-I characteristics of solar cell.
5. Magnetic field along the axis of current carrying coil – Stewart and Gee
6. Melde's Experiment.
7. Study of B-H curve of a ferro magnetic material.
8. Study of P-E loop of given ferroelectric crystal.
9. Determination of dielectric constant of a given material.
10. Determination of frequency of AC mains using sonometer.

TEXT BOOKS:

1. Engineering Physics Laboratory Manual/Observation, Physics Faculty of VNRVJIET
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S. Chand Publishers, 2017

REFERENCES:

1. Engineering Physics Lab Manual, Dr. Y. Aparna & Dr. K. Venkateswarao, V. G. S. Book links
2. Engineering Physics Practicals, Dr. B. Srinivasa Rao, V. K. V. Krishna, K. S. Rudramamba, University, Science Press, 2^{nd} Edition, Imprint of Laxmi Publications

ONLINE RESOURCES:

1. https://vlab. amrita. edu/index. php?sub=1&brch=189&sim=343&cnt=1
2. https://vlab. amrita. edu/index. php?sub=1&brch=280&sim=1518&cnt=1

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2EE102) BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Mathematics, Physics and Basic Electrical Engineering

COURSE OBJECTIVES:

- To recognize different circuit reduction techniques
- To understand the construction of electrical equipment and operation
- To practice the techniques to control and assess electrical machines and installations
- To know operation and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply different network reduction techniques to solve and analyze electrical circuits
CO-2: Realize the compatibility of electrical machines in different engineering fields
CO-3: Control different electrical machines and evaluate their performance
CO-4: Identify different basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	3	2	2	1	-	-	2	-	-	-	2	1	2
CO-2	2	3	2	2	1	-	-	2	-	-	-	2	1	2
CO-3	2	3	2	1	1	-	-	2	-	-	-	2	1	2
CO-4	2	2	2	1	1	-	-	2	-	-	-	2	1	2

LIST OF EXPERIMENTS:

1. Demonstration of safety precautions, measuring instruments, electrical and electronic components
2. Demonstration of LT switchgear components
3. Verification of KVL & KCL
4. Verification of the Superposition theorem
5. Verification of Thevenin's theorem
6. Analysis of series RL, RC, and RLC circuits
7. Load test on 1- φ transformer
8. Speed control of separately excited DC motor
9. PN Junction Diode Characteristics
10. Full Wave Rectifiers without filter
11. Transistor CE Characteristics (Input and Output)
12. Op-Amp as Inverting and Non-Inverting Amplifier

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. https://nptel. ac. in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES2AD102) FOUNDATIONS OF PYTHON PROGRAMMING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To learn about Python programming language syntax, semantics, and the runtime environment
- To be familiarized with universal computer programming concepts like data types, containers
- To be familiarized with general computer programming concepts like conditional, execution, loops & functions
- To be familiarized with general coding techniques and object-oriented programming

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the fundamental concepts of Python to develop simple programs

CO-2: Apply conditional statements, looping & built-in data structures for efficient problem-solving

CO-3: Develop modular programs, modules, and packages to improve code reusability & maintainability

CO-4: Design and implement Python using Oops principles to model real-world problems

CO-5: Implement error-handling, file i/o & basic data analysis to develop robust & data-driven applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	-	-	-	-	3	-	2	2	3
CO-2	3	2	3	2	2	-	-	-	-	-	-	2	2	3
CO-3	3	2	3	2	2	-	-	3	-	-	3	3	2	2
CO-4	3	2	3	2	2	-	-	-	-	3	-	3	2	2
CO-5	3	2	3	2	3	-	-	-	2	-	-	2	2	2

LIST OF EXPERIMENTS:

WEEK 01:

1. Install Python and configure an IDE like Jupyter, google colab.
2. Write and execute your first Python program to display "Hello, Python!".
3. Demonstrate interactive mode and script mode.
4. Use command-line arguments to run Python scripts.

WEEK 02:
1. Program to demonstrate Python's built-in data types.
2. Type conversion using int(), float(), str(), etc.
3. Calculate Simple Interest and Area of a Circle.
4. Program to swap two numbers without using a third variable.

WEEK 03:
1. Program to demonstrate all operator types.
2. Build a simple calculator using user input.
3. Program to compute the grade of a student based on input marks.
4. Program to find the largest of three numbers.

WEEK 04:
1. Check whether a number is even or odd.
2. Check whether a year is a leap year.
3. Grade calculation using if-elif-else.
4. Determine the type of a triangle (equilateral, isosceles, scalene).

WEEK 05:
1. Generate a multiplication table using for and while.
2. Program to print the Fibonacci series.
3. Factorial of a number using loops.
4. Demonstrate break, continue, and pass with examples.

WEEK 06:
1. Perform string slicing and demonstrate common string methods (upper(), lower(), replace(), etc.).
2. Check if a string is a palindrome.
3. Demonstrate list operations: append, insert, pop, reverse, sort.
4. Use list comprehensions to generate squares of numbers.

WEEK 07:
1. Create a tuple and show immutability by attempting to modify it.
2. Perform set operations: union, intersection, difference.
3. Dictionary program: add, update, delete key-value pairs.
4. Word frequency counter using a dictionary.

WEEK 08:
Lab Internal – 1

WEEK 09:
1. Demonstrate functions with different argument types: positional, keyword, default, variable-length.
2. Program to show local vs global variable behavior.
3. Recursive function to calculate factorial.
4. Recursive Fibonacci sequence generator.

WEEK 10:
1. Import and use built-in modules (math, random, datetime).
2. Create and import a user-defined module.
3. Create a simple package with multiple modules.
4. Random number guessing game using the random module.

WEEK 11:

1. Create a class Student with attributes and display student details.
2. Demonstrate constructors (__init__).
3. Implement single inheritance and multiple inheritance.
4. Demonstrate method overriding.

WEEK 12:

1. Program using try, except, else, and finally.
2. Raise and handle custom exceptions.
3. Read and write data to a text file.
4. Count the number of words and lines in a text file.

WEEK 13:

1. Validate an email ID and phone number using the re module.
2. NumPy: Create and manipulate arrays (addition, mean, transpose).
3. Pandas: Create Series and DataFrame.
4. Load CSV data using Pandas and perform basic data analysis.

WEEK 14:

Lab Internal – 2

TEXT BOOKS:

1. Fundamentals of Python First Programs, Kenneth. A. Lambert, Cengage
2. Python Programming: A Modern Approach, Vamsi Kurama, Pearson

REFERENCES:

1. Introduction to Python Programming, Gowrishankar. S, Veena A, CRC Press
2. Introduction to Programming Using Python, Y. Daniel Liang, Pearson

ONLINE RESOURCES:

1. https://www.tutorialspoint.com/python3/python_tutorial.pdf
2. https://onlinecourses.nptel.ac.in/noc21_cs78/preview
3. https://www.coursera.org/learn/python?specialization=python#syllabus

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25MN6HS103) HAPPINESS AND WELLBEING

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME			
SE-I	SE-II	SEE	TOTAL
50	50	-	100

COURSE OBJECTIVES:
- To learn sustainable strategies to develop positive attitude and happy heart
- To develop self-awareness and self-discipline to meet the needs of happiness
- To practice good health & mindfulness for wellbeing
- To adapt personality attributes of happiness and success strategies
- To nurture happiness development index for better living

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Recognize what is happiness in life and how to sustain it
CO-2: Focus on interpersonal skills for a mindful approach
CO-3: Develop mindfulness to handle challenging situations
CO-4: Recognize the importance of positive attitude for personal and professional development
CO-5: Interpret the need for nurturing happiness development index through Indicators

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
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CO-2	2	1	2	1	1	3	3	2	3	3	3	-	-	-
CO-3	2	1	2	1	1	3	3	2	3	3	3	-	-	-
CO-4	1	1	2	1	1	2	2	3	3	2	3	-	-	-
CO-5	1	1	3	1	1	3	2	3	3	2	3	-	-	-

UNIT-I:
Introduction to Happiness: Definition & theories of happiness: Hedonism theory, Desire theory, Objective list theory. Identifying potential barriers of happiness: Devaluing happiness, chasing superiority, being needy, being overly control-seeking, distrusting others, distrusting life, and ignoring the source within. Strategies for overcoming the potential barriers

UNIT-II:
Power of Emotions & Relationships: Role of emotional intelligence, self-awareness, and empathy in creating harmonious relationship with ourselves and others. Balancing emotions. Hormones that promote happiness. The importance of social connections

for happiness. Role of share & care, gratitude, forgiveness & kindness in building relationships

UNIT-III:

Health and Well-being: The link between health & happiness-exercise regularly, eat a healthy diet, get enough sleep for physical fitness. Mental wellbeing-Take notice, keep learning, stay connected with nature, and financial wellbeing. The practice of mindfulness and its benefits for mental and physical health. Moving from restlessness to restfulness- meditation and yoga to increase awareness and reduce stress

UNIT-IV:

Re-Wirement for Wellbeing: Abundance in life, freedom of choice, accepting change, ways of implementation for wellbeing: practicing habits-be proactive, begin with end-in-mind, put-first things-first, think win-win, seek first to understand then to be understood, synergize, sharpen the saw, and effectiveness to greatness

UNIT-V:

Nurturing Happiness Development Index: Exploring the sources of temporary joy and lasting happiness. Acceptance, Appreciation, forgiveness, gracefulness, and creative procrastination. Time management with four D's (delete, delay, delegate, do). Developing happiness index-track changes in happiness levels over time and identify the indicators

TEXT BOOKS:

1. The How of Happiness: A Scientific Approach to Getting the Life You Want, Sonja Lyubomirsky, Penguin Books, 2008
2. Authentic Happiness: Using the New Positive Psychology to Realize Your Potential for Lasting Fulfilment, Martin Seligman, Atria Books, 2004
3. The Book of Joy: Lasting Happiness in a Changing World, Dalai Lama, Desmond Tutu, and Douglas Abrams, Avery, 2016

REFERENCES:

1. 7-Habits of Highly Successful People, Stephen Covey, Simon & Schuster, 2020
2. Mindfulness Book of Happiness: Mindfulness and Meditation, Aimen Eman, Publish Drive Edition, 2018
3. Mindfulness at Work: How to Avoid Stress, Achieve More, and Enjoy Life, Dr. Stephen McKenzie, Exisle Publishing, 2014
4. The 8th Habit: From Effectiveness to Greatness, Stephen R. Covey, Free Press, 2004

ONLINE RESOURCES:

1. Life of Happiness and Fulfillment, Indian School of Business, Coursera https://in.coursera.org/learn/happiness
2. Science of Wellbeing, Yale University, Coursera https://www.coursera.org/learn/the-science-of-well-being